Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Section 1: Reciprocal Trigonometric Functions

1 Simplify the trigonometric expression $\frac{\sec x - \cos x}{\tan x}$.

a. $\sin x$

b. $\cos x$

c. $\cot x$

d. $\csc x$

2 Find the exact value of $\csc\left(\frac{5\pi}{6}\right)$.

a. 2

b. -2

c. $\frac{2}{\sqrt{3}}$

d. $-\frac{2}{\sqrt{3}}$

3 Which identity relates $\cot\theta$ and $\csc\theta$ for all defined angles?

a. $1 + \cot^2\theta = \csc^2\theta$

b. $1 + \csc^2\theta = \cot^2\theta$

c. $\cot^2\theta - \csc^2\theta = 1$

d. $\csc\theta = 1 + \cot\theta$

Section 2: Compound Angle Addition Formulas

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Use compound angle identities to find the exact value of $\cos(75^\circ) = \cos(45^\circ + 30^\circ)$.

a. $\frac{\sqrt{6}-\sqrt{2}}{4}$

b. $\frac{\sqrt{6}+\sqrt{2}}{4}$

c. $\frac{\sqrt{3}-1}{2\sqrt{2}}$

d. $\frac{\sqrt{2}-\sqrt{6}}{4}$

5 Simplify $\sin(x+y)\cos y - \cos(x+y)\sin y$.

a. $\sin x$

b. $\cos x$

c. $\sin(x+2y)$

d. $\cos(x-y)$

6 If $\tan A = 2$ and $\tan B = 3$, what is $\tan(A+B)$?

a. -1

b. 1

c. 5

d. $\frac{5}{7}$

Section 3: Double Angle Formulas

7 If $\sin\theta = \frac{3}{5}$ and θ is acute, what is $\cos(2\theta)$?

a. $\frac{7}{25}$

b. $\frac{24}{25}$

c. $-\frac{7}{25}$

d. $\frac{16}{25}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Under the same conditions ($\sin\theta = \frac{3}{5}$ acute), what is $\sin(2\theta)$?

a. $\frac{24}{25}$

b. $\frac{7}{25}$

c. $\frac{12}{25}$

d. $\frac{6}{5}$

9 Express $\cos^2 x$ in terms of a first-power cosine function.

a. $\frac{1+\cos(2x)}{2}$

b. $\frac{1-\cos(2x)}{2}$

c. $\frac{1+\sin(2x)}{2}$

d. $1-\sin^2 x$

Section 4: Linear Trigonometric Combinations (R-form)

10 Express $3\sin x + 4\cos x$ in the form $R\sin(x+\alpha)$ with $R>0$ and $0<\alpha<\frac{\pi}{2}$.

a. $5\sin(x+0.927)$

b. $5\sin(x+0.643)$

c. $7\sin(x+0.927)$

d. $25\sin(x+0.927)$

11 What is the maximum value of the function $f(x) = 5\cos x - 12\sin x + 3$?

a. 16

b. 13

c. 10

d. 20

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 Solve $\sin(2x) = \cos x$ for $x \in [0, 2\pi)$.

a. $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$

b. $x = \frac{\pi}{6}, \frac{5\pi}{6}$

c. $x = 0, \pi$

d. $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647